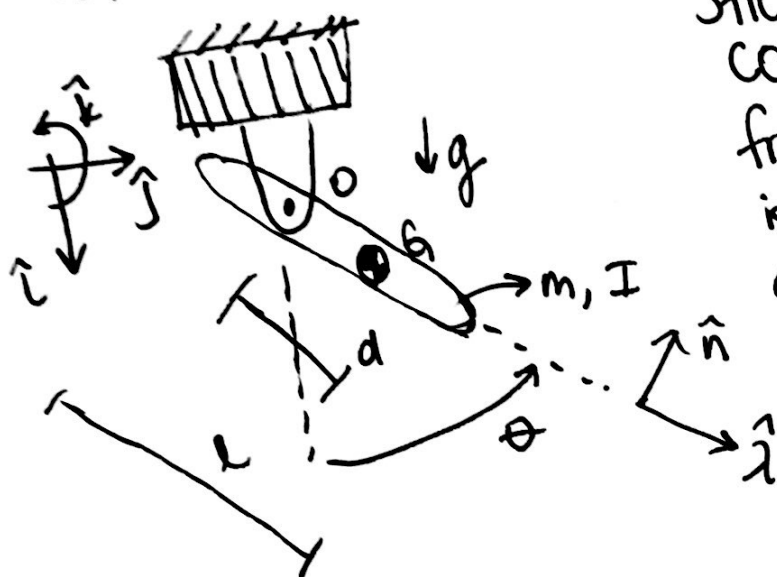


Old Exam: FA2012

Given:



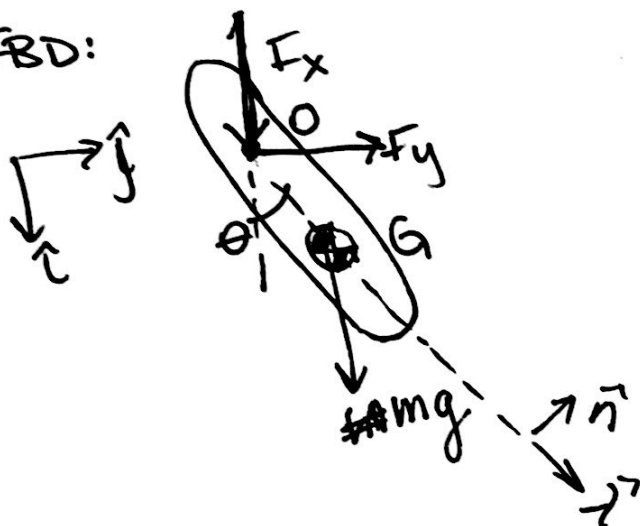
Stick mounted to a hinge.
COG is distance, d , away from hinge location. Hinge is fixed. All other parameters are given in diagram.

- i) Find eom
- ii) Find $\vec{F}$
- iii) Find d to minimize small θ oscillations

$$\hat{\lambda} = \cos\theta \hat{i} + \sin\theta \hat{j}$$

$$\hat{n} = -\sin\theta \hat{i} + \cos\theta \hat{j}$$

FBD:



i) EOM:

Angular Momentum Balance: $\sum \vec{M}_{/O} = \sum \vec{H}_{/O}$

$$-mgd \sin\theta \hat{k} = d \hat{\lambda} \times m(d\ddot{\theta} \hat{n} - d\dot{\theta}^2 \hat{\lambda}) + I\ddot{\theta} \hat{k}$$

$$-mgd \sin\theta \hat{k} = (md^2\ddot{\theta} + I\ddot{\theta}) \hat{k}$$

$$\boxed{\ddot{\theta} = \frac{mgd \sin\theta}{md^2 + I}}$$

(1)

ii) Find $\vec{F}$

Linear Momentum Balance $\Sigma \vec{F} = \cancel{m\vec{a}} m\vec{a}$

$$\vec{F} + mg\hat{c} = m(d\ddot{\theta}\hat{n} - d\dot{\theta}^2\hat{\lambda})$$
$$\boxed{\vec{F} = -mg\hat{c} + \frac{m^2 d^2 g}{md^2 + I} \sin\theta \hat{n} - md\dot{\theta}^2 \hat{\lambda}}$$

iii) Minimize small angle oscillations

We know that $\omega^2 = \frac{mgd}{md^2 + I}$ from small angle approximation

We want to minimize ω ; $\frac{d\omega^2}{dd} = 0$ allows us to solve for this optimization. Using quotient rule:

$$0 = \frac{mg(md^2 + I) - mgd(2md)}{(md^2 + I)^2}$$

$$0 = m^2 g d^2 + mgI - 2m^2 g d^2$$

$$md^2 = I$$

$$\boxed{d = \sqrt{I/m}}$$

```
% FALL 2012 Exam Question
% MATLAB code for the simulation of the motion of the stick mounted at the
% hinge
% Author: Deepthi Krovvidi (dk562)
```

```
% Define System Parameters
disp('Defining system parameters')
p.m = 3;
p.I = 2;
p.g = 1;
p.l = 2;
p.lo = 0.3;
p.log = 0.2;
```

```
% Define Initial Conditions
th = pi/3;
thd = 0;
p.ic = [th thd];
```

```
% Define time step and vector
timeinterval = 0.05;
totaltime = 30;
p.t = linspace(0,totaltime,(totaltime/timeinterval)+1);
```

```
% Define options
options = odeset('abstol', 1e-6, 'reltol', 1e-6);
```

```
% Ensure that the given parameter lengths are reasonable such that the
% length from one end to the hinge plus the hinge to the center of gravity
% are less than the total length of the rod
disp('Checking Initial Conditions')
if p.lo+p.log > p.l
    error('Geometry is Invalid')
    quit
end
```

```
% If the above check passes, we continue to simulate the motion using ode45
% to numerically integrate
disp('Using ODE45 to numerically solve for theta')
[t, z] = ode45(@HINGE_RHS, p.t, p.ic, options, p);
```

```
% Now we can plot the animation
disp('Plotting motion of the rod')
HingeAnimate(t, z, p.l, p.lo)
```

```
% The RHS function for numerically solving the equations of motion
% Deepthi Krovvidi (dk562)
```

```
function [mdot] = HINGE_RHS(t, m_init, p)
```

```
% Breaking out parameters
m = p.m; g = p.g; d = p.log; I = p.I;
```

```

% Breaking out variables
th = m_init(1);
thd = m_init(2);

% Solve for equations of motion
thdot = thd;
thddot = (m*-g*d*sin(th))/(m*d^2 + I);

mdot = [thdot thddot]';
end

```

Hinge Animation

% This function is made by Professor Andy Ruina - the original is called doublepenanimate.m and uses PLOT and SET functions
 % It has been modified to plot the motion of a hinge
 % Deepthi Krovvidi (dk562)

```

function [] = HingeAnimate(t, z, l, lo)
clf

n=length(t); dur=t(end);

% reference before translation and rotation
bar1ref = [0 lo;0 0];
bar2ref = [0 -(l-lo);0 0];

hinge=plot(0,0, 'marker','.', 'markersize',[50], ...
    'color','black');hold on;
title('Motion of System')
xlabel('X Coordinate')
ylabel('Y Coordinate')
axis('equal')
axis([-3 3 -3 3]);

bar1pic=plot(bar1ref(1,:),bar1ref(2,:), ...
    'linewidth', 3, 'color','green');
bar2pic=plot(bar2ref(1,:),bar1ref(2,:), ... % +l1 for init disp
    'linewidth', 3, 'color','red');

%set(bar1pic,'linewidth',7) % fatten bar 1 a bit

%Wallpaper
%Good to keep this outside of the plotting loop
%points = 4*rand(2,400)-2;
%background = plot(points(1,:),points(2:),'bo','markersize',20 )

pause(1)% Hold still for a second before starting

for i=1:n
    pause(.01) %Use to control the speed of the animation

```

```


 R1=[cos(th1) -sin(th1)    ; sin(th1) cos(th1)]; % one rotation matrix | bar1=R1*bar1ref; | bar2=R1*bar2ref; | endpt(i,:) = [bar2(2,end), bar2(1,end)]; | set(bar1pic,'xdata',bar1(2,:),'ydata',bar1(1,:)); | set(bar2pic,'xdata',bar2(2,:), ... | 'ydata',bar2(1,:)    ); |

end
plot(endpt(:,1), endpt(:,2));
end

```

System Output:

```

Defining system parameters
Checking Initial Conditions
Using ODE45 to numerically solve for theta
Plotting motion of the rod

```

